

**Study of Binary Fission in *Amoeba* and Budding in *Hydra*****Aim**

To study (1) Binary fission in *Amoeba* and (2) Budding in *Hydra* with the help of permanent slides.

**Requirements**

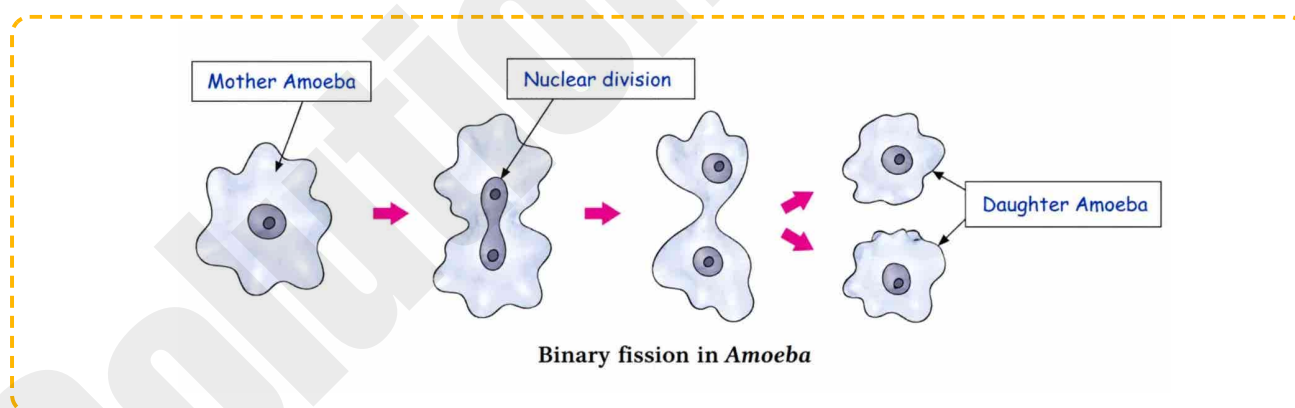
Slides of binary fission in *Amoeba* and budding in *Hydra*, and a compound microscope.

**Procedure**

1. Observe the slides showing binary fission in *Amoeba* and budding in *Hydra* under a compound microscope.
2. Set up the mirror and the objective lens of the compound microscope.
3. First, focus the slides using the low power objective lens, switching to the high power objective lens if needed, and note down your observations.

**1. Binary Fission in *Amoeba*****Diagram**

(Label the different parts in the given diagram)

**Observations: Strike out the incorrect word**

1. Binary fission of *Amoeba* starts with the division of nucleus/~~cell wall~~.
2. Binary fission is type of mitotic/~~meiotic~~ cell division.
3. By binary fission one *Amoeba* gives rise to two/~~four~~ daughter Amoebae.

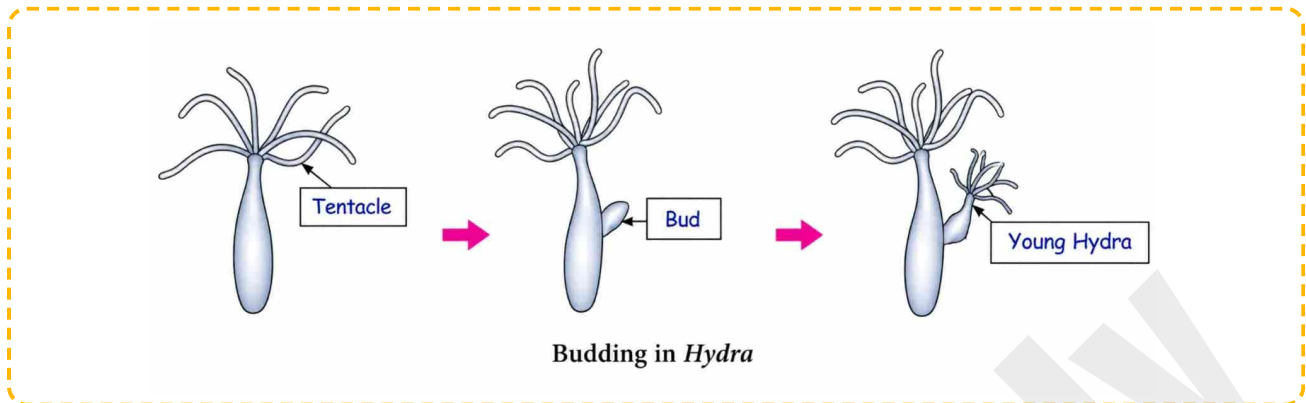
**Inference**

Binary fission and budding are types of asexual reproduction.

## 2. Budding in *Hydra*

### Diagram

(Label the different parts in the given diagram)



### Observations: Strike out the incorrect word

1. During reproduction, outgrowth developed on body of *Hydra* is called bud/~~tentacle~~.
2. The bud of *Hydra* is multicellular/~~unicellular~~.
3. Young *Hydra* ~~remain attached on~~/detaches from parent *Hydra* and starts independent life.

### Inference

Both binary fission and budding are asexual modes of reproduction.

### Multiple Choice Questions

1. For observation of the *Hydra* bud clearly, first focus under the low power lens and then under the high power lens in order to observe .....

- (A) the minute part in less area (B) the wide area  
(C) more buds (D) all of these

**Answer: (A) the minute part in less area**

2. Which stain will clearly show *Amoeba* and *Paramecium* in a drop of water taken on a slide and observed under a microscope?

- (A) Methylene blue (B) Iodine  
(C) Safranin (D) Eosin

**Answer: (A) Methylene blue**

3. What is the type of reproduction in *Hydra*?

- (A) Binary fission (B) Budding  
(C) Multiple fission (D) None of these

**Answer: (B) Budding**

**4. A slide of *Amoeba* with elongated nuclei represents which of the following?**

- (A) Budding (B) Multiple fission  
(C) Binary fission (D) Regeneration

**Answer: (C) Binary fission**

**5. Which of the following is a unicellular fungus that shows budding?**

- (A) Mucor (B) Yeast  
(C) Amoeba (D) None of these

**Answer: (B) Yeast**

**6. Which of the following is a mode of asexual reproduction?**

- (A) Fission (B) Budding  
(C) Spore formation (D) All of these

**Answer: (D) All of these**

### **Remember**

1. Binary fission is usually performed by living organisms during favourable conditions i.e. availability of abundant food material.
2. Asexual reproduction occurs by budding in yeast – a unicellular fungus.

Date: \_\_\_\_\_

Teacher's Signature: \_\_\_\_\_